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Problem: Analyze $f(x) = x^3 - 3x^2 + 2$

Solution:

Step 1: First Derivative

$$f'(x) = 3x^2 - 6x.$$

Using the power rule, the first derivative of x^3 is $3x^2$, and the derivative of $-3x^2$ is $-6x$. The constant

Step 2: Critical Points

$$\text{Set } f'(x) = 0:$$

$$3x(x - 2) = 0$$

Thus, the critical points are $x = 0$ and $x = 2$.

Step 3: Second Derivative Test

$$f''(x) = 6x - 6.$$

At $x = 0$: $f''(0) = -6 < 0$, which means there is a local maximum at $x = 0$.

At $x = 2$: $f''(2) = 6 > 0$, which means there is a local minimum at $x = 2$.